

Number-Based Problems

1. Check if a number is prime
2. Check if a number is Palindrome (e.g., 121, 12321)
3. Reverse a number (e.g., 123 $\rightarrow$ 321)
4. Find the Factorial of a number
5. Generate Fibonacci series up to n terms
6. Check if a number is Armstrong (e.g., $153 = 1^3 + 5^3 + 3^3$)
7. Find GCD and LCM of two numbers
8. Swap two numbers without using a third variable
9. Check if a number is Perfect Number (e.g., $6 = 1 + 2 + 3$)
10. Count digits in a number

String-Based Problems

11. Check if a String is Palindrome (e.g., "madam")
12. Reverse a String (with and without built-in functions)
13. Count vowels and consonants in a string
14. Check if two Strings are Anagrams (e.g., "listen" and "silent")
15. Find duplicate characters in a string
16. Remove duplicate characters from a string
17. Count occurrence of each character in a string
18. Find the first non-repeating character in a string

Array-Based Problems

19. Find the largest and smallest element in an array
20. Reverse an array
21. Find duplicate elements in an array
22. Sort an array (Bubble sort, Selection sort - manually)
23. Find the second largest element in an array
24. Merge two sorted arrays
25. Find missing number in an array of 1 to n
26. Remove duplicates from an array
27. Rotate an array by k positions

Pattern Printing Problems

28. Print a pyramid/triangle pattern of stars
29. Print a diamond pattern
30. Print Pascal's Triangle

Logic-Based / Others

31. Check leap year
32. Find sum of digits of a number

- 33. **Bubble sort / Selection sort implementation**
- 34. **Binary search implementation**
- 35. **Linear search implementation**
- 36. **Check if a String/Number is a valid Palindrome** using two-pointer approach
- 37. **FizzBuzz problem** (print Fizz for multiples of 3, Buzz for 5, FizzBuzz for both)